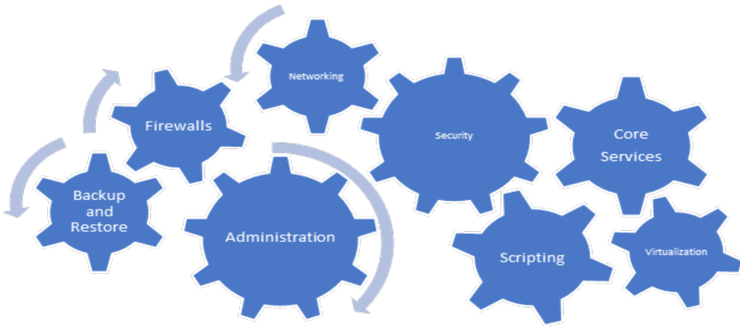


System Administration

Lab Environment – Second Half

Overview:

This document describes the lab environment you'll be working in for the second half of the course.

Pre-requisites:

You will need the following items to create this setup

- Ubuntu 26.04 Server ISO
- Windows Server 2022 ISO
- A virtualization platform of your choice (VirtualBox, VMWare, KVM, etc.). Note that your professor will not provide tech support/troubleshooting for your chosen virtualization platform.

Objectives and Tools:

- Create a daisy-chain network that consists of three (3) VMs spread across two distinct networks.
- Demonstrate manual configuration of static routing on Linux OS.

Details:

Networks:

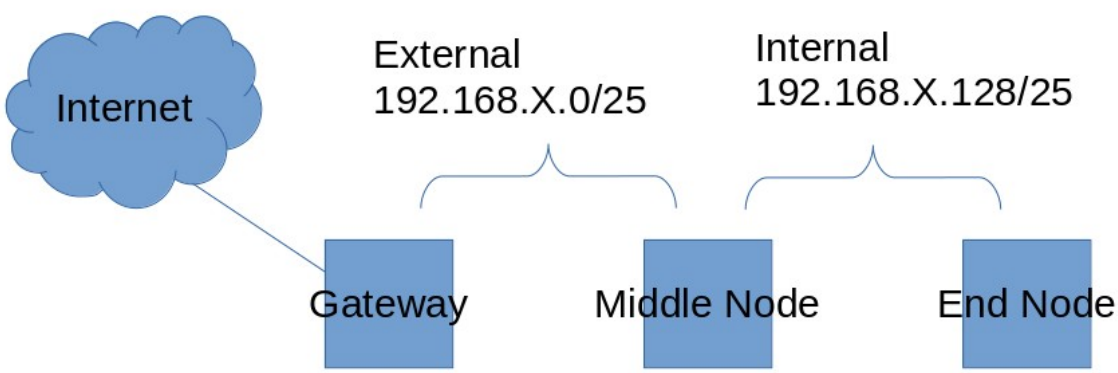
- You have been given a network number on blackboard (see the Network Number column in the grade centre). This number is third octet of your assigned network space (192.168.X.0/24). You will sub-divide this address space to create separate networks.
- Note: Virtualization and cloud platforms automatically claim the first several addresses in any network created in them. Do not use any of the first three addresses in either network.
- Create two distinct networks in your chosen virtualization platform. Note: These must be created as **two separate networks** in your virtualization platform. Attempting to use a single network and just assign addresses will cause some services to function incorrectly.
    - Internal – 192.168.X.128/25: Any services or information considered sensitive will be placed in this subnet. No direct access from the outside world will be permitted. Any traffic attempting to reach other networks must pass through your node1 machine (see below).
    - External – 192.168.X.0/25: While called 'external' this network will also not provide direct access to the outside world. It is simply the outermost layer of your network setup. Any traffic attempting to reach the outside world (e.g. the internet) will pass through your gateway machine (see below) to be forwarded to a network that actually has outside access.
- You may assign your addresses statically, but are encouraged to use DHCP in order to get more practice with it.

Gateway:
- OS: Ubuntu 26 LTS
- Hostname: gateway
- 2 Interfaces
  - 1 in the external network (192.168.X.5/25)
  - 1 DHCP-assigned NAT address provided by your virtualization platform. Must be capable of reaching the internet.
- Note this is the only machine that should have direct access to the class network (or anything else outside your lab and assignment networks). All traffic passing between your other VMs and the outside world must pass through this machine. Machines outside your network may respond to traffic your machines send, but must not be able to **initiate** connections to your machines.

Middle Node:
- OS: Windows Server 2022
- Hostname: node1
- 2 Interfaces
  - 1 in the external network (192.168.X.6/25)
  - 1 in the internal network (192.168.X.132/25)
- This machine must be able to reach the outside world by passing traffic to your gateway machine (e.g. to obtain updates) and will act as a gateway to allow traffic from the internal network to reach the external one (and potentially the outside world).

End Node
- OS: Ubuntu 26 LTS
- Hostname: node2
- 1 Interface in the internal network (192.168.X.133/25)
- This machine must be able to reach the outside world by passing traffic to your middle node (e.g. to obtain updates).

Network Diagram:



General Guidelines

- *All machines have reboot-persistent ip addresses and* routes
- *All machines are able to reach the internet using ip addresses and FQDNs*
- *All machines are able to communicate with each other using ip addresses*
- *Submissions with deactivated security features (e.g. no firewall, firewall permitting all traffic, AppArmor disabled or inactive) will not be accepted as complete.*
    - *Security features exist to protect your machines. You are in an IT security program. Being able to work with these features to protect your machines, networks, and the data on them is a basic expectation of this field.*
    - *The Ubuntu machines will use nftables as their firewall (using the inet filter table, with the gateway machine also using the postrouting chain of the nat table to masquerade traffic leaving your network).*
    - *The Windows server will use the public zone of the Windows Defender Firewall.*